Smart Offer Engine: Progress Summary

Hackathon project · decides which offer each e-commerce customer should get, checks every AI choice against business rules, and measures whether offers make money · status as of 22 Oct 2026

Customer data 200 synthetic, 8 personas › **Features +** segment › **Rule pre-filter** allowed offers + ranges › **Claude agent** profit method › **Rule checker** R1–R18 › **Retry once,** else rules fallback › **Offer log +** preview page

Results: simulated policy comparison (200 customers, 1,000 simulated weeks, per week)

Policy	Messaged	Extra purchases	Discount cost	Paid to buy-anyway	Extra net margin
No offers	0	0	₹0	₹0	₹0
10% to everyone	185	+23.7	₹12,223	₹8,421	**−₹1,045**
Our rules	155	+19.5	₹3,237	₹1,645	**+₹7,231**

A blanket 10% buys the most extra purchases but 69% of its cost goes to customers who would have bought anyway, so it earns less than sending nothing. The rules get ~80% of the extra purchases for ~26% of the cost. Response behaviour per persona is an assumption (config/response_model.yaml); the AI policy is not yet evaluated.

What is built

- **Synthetic data**: 200 customers from 8 hidden personas with ~5,000 orders, ~3,000 sessions, cart events, points that expire after 12 months; features computed from raw tables; 89% land in the matching segment.
- **Offer catalog**: 10 offers (reminders, points, free shipping, flat ₹, %, category deal, welcome, early access).
- **Rule engine + checker**: every offer, from AI or rules, is validated before it is logged.
- **Claude agent** (claude-opus-5-5, JSON output): system message holds the full rules, catalog and a profit method; customer data and per-offer economics go in the user turn as data.
- **Fallback**: per-segment offer ladders and value strategy; used when the AI fails, times out or breaks rules twice.
- **Offer log** (SQLite): offer, message, cost, reason, decided by AI / rules / system, failure reason, violations, flag.
- **Preview page** (Flask): every customer's offer, who decided and why, simulated time, budget, flags.
- **Evaluation** (evaluate.py): Monte Carlo comparison of policies plus a holdout-test check.
- **Calibration code** for Flipkart, RetailRocket and REES46 data (waiting for Kaggle downloads).
- **Tests**: 8 passing (rules, welcome scope, invented terms, agent OK / retry / rejected / timeout, frequency caps).

Business rules (config/rules.yaml)

- R1 consent · R2 offer allowed for segment · R3 catalog range · R4 segment caps · R5 margin floor 10% · R6 flat ≤ 15% of avg order · R8 one active offer · R9 ≥24h apart, ≤3 per 7 days · R10 14-day discount cooldown · R11 quiet hours 22:00–08:00 · R12 cart reminder before paid offers · R13 daily budget ₹20,000 · R15 message: ≤300 chars, true value, no pressure, no invented terms · R16 no sensitive attributes · R17 valid JSON, catalog offer only · R18 15% control group.

Findings and fixes

- **Budget went first-come-first-served**: now spent on highest value at stake first.
- **Simulation flaw**: unredeemable min-order discounts converted for free; fixed (buyers top up the basket).
- **Tuning**: flat ₹ first for DORMANT, max values for DEAL_SEEKER: +₹6,572 → +₹7,010/week, stable on 3 seeds.
- **Category-limited welcome offer**: new customers browsing Mobiles were blocked by the margin rule; the offer now applies only to Fashion, Home, Beauty, Books. NEW beats the blanket (+₹31 vs +₹30 per customer); total +₹7,231.
- **Kept margin floor and cooldown**: loosening them gains only ~₹250/week and risks thin margins and coupon habits.
- **Live AI test** (9 customers, 1 per segment): 8 AI decisions, all passed the rule checker, 0 flags, 4–7 s each. Loyal/VIP get free perks, at-risk get points, buy-anyway customers get nothing.
- **Invented terms**: AI wrote promises like "cart saved for 3 days"; added an R15 check (days, ₹, %, unsupported claims must match the catalog) and gave the AI an explicit facts list. Rerun: none found.
- **Measurement lesson**: one week with 165 vs 35 control customers measures ≈ +22% ± 12% uplift when the truth is ≈ +9.5%; real tests need more customers or weeks.

Next steps

- Run the AI policy evaluation (~200 API calls) and compare AI vs rules on profit.
- Delivery channel: Discord bot (free, real buttons) for classmates; log Redeem / Not interested / Stop.
- Calibrate personas with Kaggle data and the response model with the Hillstrom campaign data.
- Speed up AI decisions (6–9 s each) for batch runs.

Example AI decisions from the live run (all passed the rule checker)

Customer	AI choice	Message sent (excerpt)	AI's reason (excerpt)
Harpreet · NEW	Welcome 10%	Enjoy 10% off your first order of ₹499 or more on Beauty, Fashion, Home and Books. Valid for 7 days.	No orders, 12 browses without buying; ≈ ₹47 expected profit vs ₹45 for free delivery.
Riya · CART_ABANDONER	Cart reminder	Your Perfume is still waiting in your cart. Come back whenever you're ready.	Free nudge with real lift; paid offers not allowed yet (R12).
Deepak · AT_RISK	200 bonus points	200 bonus points on your next order… valid for 7 days.	Rarely uses coupons, so a discount mostly pays buy-anyway orders; points ≈ +₹14 vs a loss for 10% off.
Sai · DEAL_SEEKER	No offer	-	Not drifting, would buy anyway (0.59): 5% gains ≈ ₹9.5 but costs ≈ ₹21.